DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 5: Risk Aversion Coefficients & Certainty Equivalence

Certainty Equivalent · Risk Premium · Arrow–Pratt Measures

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Today's Agenda

1. Certainty equivalent and risk premium: definitions
2. Deriving the Arrow–Pratt approximation
3. Absolute vs. relative risk aversion
4. DARA / CARA / IARA – and why it matters

Learning Objectives

- Define and compute certainty equivalent and risk premium.
- Derive $A(W)$ via Taylor expansion.
- Approximate a risk premium using variance and $A(W)$.
- Distinguish DARA/CARA/IARA and interpret economically.

Two Precise Numbers

Definition: Certainty Equivalent

CE solves $u(CE) = \mathbb{E}[u(W)]$.

Definition: Risk Premium

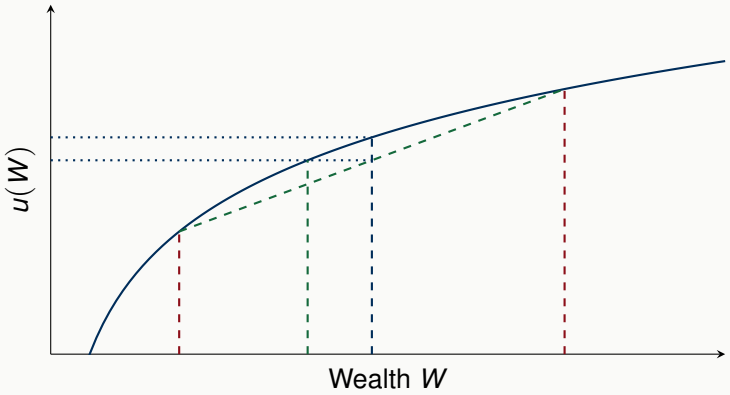
$$\pi = \mathbb{E}[W] - CE$$

The Risk Premium is an Insurance Premium

Fire Insurance

Your home: 1% chance of Rs. 10,00,000 fire damage. Expected loss: Rs. 10,000. A risk-neutral insurer breaks even charging exactly that. But *you*, risk-averse, will happily pay *more* – eliminating the small chance of a devastating loss is worth extra. That "extra" is π.

CE and π on the Utility Curve



Setting Up the Taylor Expansion

Let $W = \overline{W} + \varepsilon$, $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$.

Step 1: Expand $u(W)$ around $\overline{W}$ (2nd order)

$$\mathbb{E}[u(W)] \approx u(\overline{W}) + \tfrac{1}{2} u''(\overline{W})\sigma^2$$

Step 2: Expand $u(CE) = u(\overline{W} - \pi)$ (1st order)

$$u(CE) \approx u(\overline{W}) - \pi\, u'(\overline{W})$$

Step 3: Equate and Solve

$$u(\overline{W}) - \pi u'(\overline{W}) = u(\overline{W}) + \tfrac{1}{2}u''(\overline{W})\sigma^2$$

$$\pi \approx -\frac{1}{2}\frac{u''(\overline{W})}{u'(\overline{W})}\sigma^2$$

Arrow–Pratt Absolute Risk Aversion

$$A(W) = -\frac{u''(W)}{u'(W)} \qquad \Longrightarrow \qquad \pi \approx \tfrac{1}{2}A(\overline{W})\sigma^2$$

Why the Minus Sign? Why Divide by u'?

Remark

$u'' < 0$ for risk-averse $\Rightarrow$ minus sign makes $A(W) > 0$. Dividing by $u'(W)$ **normalizes away** arbitrary units of utility – multiply u by any positive constant, $A(W)$ is unchanged, exactly as it should be.

Worked Example

$u(W) = \ln(W)$, so $A(W) = 1/W$

At $\overline{W} =$ Rs. 10,00,000, risk with $\sigma^2 = (\text{Rs. } 1,00,000)^2$:

$$\pi \approx \tfrac{1}{2} \cdot \frac{1}{10{,}00{,}000} \cdot (1{,}00{,}000)^2 = \tfrac{1}{2} \cdot 10{,}000 = \text{Rs. } \mathbf{5{,}000}$$

Proportional Risks Need a Different Tool

Definition: Relative Risk Aversion

$$R(W) = W \cdot A(W) = -\frac{W\,u''(W)}{u'(W)}$$

For proportional risk $W = \overline{W}(1+\tilde{\varepsilon})$, $\mathrm{Var}(\tilde{\varepsilon}) = \sigma_{\%}^2$:

$$\frac{\pi}{\overline{W}} \approx \tfrac{1}{2} R(\overline{W})\sigma_{\%}^2$$

ARA vs. RRA: Two Different Questions

Fixed Rupees vs. Fixed Percentage

Ask a person with Rs. 10,000 and a billionaire: "risk a fixed Rs. 1,000 for a 50% shot at Rs. 2,200?" Answers may differ wildly – that's **absolute** risk aversion $A(W)$. Now ask: "risk 10% of your wealth for a 50% shot at $+22\%$?" Both might answer identically – that's **relative** risk aversion $R(W)$.

How Risk Aversion Changes with Wealth

- **DARA** (Decreasing $A(W)$): richer $\rightarrow$ willing to risk *more rupees*. Empirically dominant.
- **CARA** (Constant $A(W)$): same rupee risk at every wealth level. Mathematically convenient.
- **IARA** (Increasing $A(W)$): richer $\rightarrow$ smaller rupee risks. Rarely used – contradicts evidence.

An analogous DRRA/CRRA/IRRA classification applies to $R(W)$.

Key Takeaways

Key Takeaway

- CE solves $u(CE) = \mathbb{E}[u(W)]$; $\pi = \mathbb{E}[W] - CE$ – exactly what an insurance premium represents.
- **Arrow–Pratt**: $\pi \approx \frac{1}{2} A(\overline{W})\sigma^2$, with $A(W) = -u''/u'$.
- For proportional risk, use $R(W) = W \cdot A(W)$ instead.
- Empirically: $A(W)$ tends to **decrease** with wealth (DARA); $R(W)$ is often modeled as roughly **constant** (CRRA).

Check Your Understanding

Before next lecture

1. For $u(W) = -e^{-aW}$, derive $A(W)$ and $R(W)$. Which is constant?
2. An insurer's fair cost is Rs. 8,000; policyholder has $A(\overline{W}) = 0.0002$, $\sigma =$ Rs. 50,000. Estimate max acceptable premium.
3. Why is $A(W)$ right for a fixed-rupee deductible, but $R(W)$ right for an equity allocation decision?

Looking Ahead

We have $A(W)$ and $R(W)$ – generic to *any* $u(\cdot)$.

Which *specific* $u(W)$ gives clean, constant DARA/CRRA or CARA?

Log vs. Power Utility – Lecture 6